1.

Let p(x, y) represents the statement “x loves y” over the domain of all people in the world, and a constant O that represents “Oliver”. Write the statement as well-formed formulas in predicate logic.

a. “There is exactly one person whom everybody loves.”

$\exists! y. \forall x. p(x, y)$

b. “There are exactly two people whom Oliver loves.”

$\exists! y. (\exists! x. (p(O, x) \land p(O, y)))$

c. “There is somebody who loves nobody but themselves.”

$\exists x \forall y (p(x, x) \land \neg p(x, y))$

2.

Prove the following claim using the direct proof strategy.

Claim: For any even integer n, n^2 is even.

Let n be an arbitrarily even integer.

$n = 2k, k \in \mathbb{Z}$

$n^2 = 4k^2$

$n^2 = 2(2k^2)$

Hence, n^2 is even.

Prove the following claim in the integer domain using the indirect proof strategy.

Claim: If ab is even, then a is even or b is even.

Let ab be an even number, such that both a and b are odd.

$a = 2k + 1, b = 2m + 1, k, m \in \mathbb{Z}$

$(2k + 1)(2m + 1) = 4km + 2k + 2m + 1$

$2(2km + m + k) + 1 \equiv 2p + 1, p \in \mathbb{Z}$

$2p + 1 \not\equiv 2n, n \in \mathbb{Z}$

Hence, a or b are even.

Prove the following claim in the integer domain using the indirect proof strategy.

Claim: For any integer n, if $n^3 + 5$ is odd, then n is even.

Let $n^3 + 5$ be odd, where n is odd.

$n = 2k + 1, k \in \mathbb{Z}$

$(2k + 1)^3 + 5 = 8k^3 + 12k^2 + 6k + 6$

$2(k^3 + 6k^2 + 3k + 3) \equiv 2m, m \in \mathbb{Z}$

Contradiction, hence n is even.

Prove the following claim in the integer domain using proof by contradiction.

Claim: If ab is odd, then a is odd or b is odd.

Let ab be odd, where a and b are even.

$a = 2k, b = 2m, k, m \in \mathbb{Z}$

$(2k)(2m) = 4km$

$2(2km) \equiv 2p, p \in \mathbb{Z}$

Contradiction, hence a and b are odd.

Prove the following claim in the integer domain using case distinction.

Claim: $n(n+1)$ is always even.

Case 1: Let n be even.

$n = 2k, k \in \mathbb{Z}$

$2k(2k+1) = 4k^2 + 2k$

$2(2k^2 + k) \equiv 2m, m \in \mathbb{Z}$

Hence, for when n is even, $n(n+1)$ is even.

Case 2: Let n be odd.

$n = 2k + 1, k \in \mathbb{Z}$

$(2k+1)(2k+2) = 4k^2 + 6k + 2$

$2(2k^2 + 3k + 1) \equiv 2m, m \in \mathbb{Z}$

Hence, for when n is odd, $n(n+1)$ is even.

Therefore, $n(n+1)$ is always even.

3.

For the following statements, decide whether they are true or false. Prove the true statements, picking your own proof strategy. For the false statements, find a counterexample.

a.

Claim: The sum of an even integer and an odd integer is an odd integer.

Let a and b be even and odd integers respectively.

$a = 2k, b = 2m + 1, k, m \in \mathbb{Z}$

$2k + (2m + 1) = 2k + 2m + 1$

$2(k + m) + 1 \equiv 2p + 1, p \in \mathbb{Z}$

Hence, the sum of even and odd integers is odd.

b.

Claim: If $x + y$ is even, then x and y have the same parity *(parity is the property of being odd or even)*

Case 1: Let x and y be even.

$x = 2k, y = 2m, k, m \in \mathbb{Z}$

$2k + 2m$

$2(k + m) \equiv 2p, p \in \mathbb{Z}$

Hence, when x and y are even, $x + y$ is even.

Case 2: Let x and y be odd.

$x = 2k + 1, y = 2m + 1, k, m \in \mathbb{Z}$

$(2k + 1) + (2m + 1) = 2k + 2m + 2$

$2(k + m + 1) \equiv 2p, p \in \mathbb{Z}$

Hence, when x and y are odd, $x + y$ is even.

Case 3: Let x and y be even and odd respectively.

$x = 2k, y = 2m + 1, k, m \in \mathbb{Z}$

$2k + 2m + 1$

$2(k + m) + 1 \equiv 2p + 1, p \in \mathbb{Z}$

Hence, when x is even and y is odd, $x + y$ is not even.

Case 4: Let x and y be odd and even respectively.

$x = 2k + 1, y = 2m, k, m \in \mathbb{Z}$

$2k + 1 + 2m$

$2(k + m) + 1 \equiv 2p + 1, p \in \mathbb{Z}$

Hence, when x is odd and y is even, $x + y$ is not even.

Therefore, claim is $\top$

Claim: All primes are odd.

Let $n = 2$

$2 = 2(1)$, thus 2 is even.

However, 2 is a prime number, only divisible by 1 and itself.

Therefore, claim is $\bot$

Claim: For all integers n, the term $n^3 - n$ is even.

Case 1: Let n be even.

$n = 2m, m \in \mathbb{Z}$

$(2m)^3 - 2m = 8m^3 - 2m$

$2(4m^3 - m) \equiv 2p, p \in \mathbb{Z}$

Hence, when n is even, $n^3 - n$ is even.

Case 2: Let n be odd.

$n = 2m + 1, m \in \mathbb{Z}$

$(2m+1)^3 - (2m+1) = 8m^3 + 12m^2 + 4m$

$2(4m^3 + 6m^2 + 2m) \equiv 2p, p \in \mathbb{Z}$

Hence, when n is odd, $n^3 - n$ is even.

Therefore, claim is $\top$

4.

Prove each of the following existential statements by providing a witness. State the witness explicitly and verify that it satisfies the predicate.

a. $\exists x : \mathbb{Z}.\ x^2 = x$

$x = 1$

$1^2 = 1$

b. $\exists x : \mathbb{Z}.\ x + 1 > x$

$x = 3$

$3 + 1 = 4$

c. $\exists x : \mathbb{Z},\ x^2 > x$

$x = -5$

$(-5)^2 = 25$

5.

For each statement below, decide whether to disprove it (find a counterexample) or prove it. If disproving, give an explicit counterexample and explain why it works. If proving, use the arbitrary element strategy.

a. $\forall x : \mathbb{Z}.\ x^2 > x$

Let $x = 1$

$x^2 = 1^2 = 1 \not> 1$

Thus, statement is $\bot$

b. $\forall x : \mathbb{R}.\ x^2 \geq 0$

Let x be an arbitrary real number

Case 1: $x < 0$

Any negative real multiplied by itself is positive

Case 2: $x > 0$

Any positive real multiplied by itself is positive

Case 3: $x = 0$

Any zero-value multiplied by itself is zero-value

Therefore, statement is $\top$

c. $\forall x : \mathbb{Z}.\ x^2 \geq x$

Let x be an arbitrary integer

Case 1: $x > 0$

Any positive integer squared is bigger than the integer

Case 2: $x < 0$

Any negative integer squared is bigger than the integer

Case 3: $x = 0$

Any zero-value is equal to zero

Therefore, statement is $\top$

6.

Write a formal natural deduction proof of the following claims. Use the rules $\forall$ introduction, $\forall$ elimination, $\Rightarrow$ introduction, and $\Rightarrow$ elimination, stating which lines each step depends on.

Claim: Given $\forall x.\ (S(x) \Rightarrow W(x))$, and $\forall x.\ (W(x) \Rightarrow P(x))$, prove that $\forall x.\ (S(x) \Rightarrow P(x))$, where $S(x)$: x is a student; $W(x)$: x works hard; $P(x)$: x passes.

Let x be an arbitrary student.

1	$\forall x.\ (S(x) \Rightarrow W(x))$	*Premise*
2	$\forall x.\ (W(x) \Rightarrow P(x))$	*Premise*
3	$S(x)$	*Assumption*
4	$S(x) \Rightarrow W(x)$	$\forall E, [1]$
5	$W(x)$	$\Rightarrow E, [1]$
6	$W(x) \Rightarrow P(x)$	$\forall E, [2]$
7	$P(x)$	$\Rightarrow E, [2]$
8	$S(x) \Rightarrow P(x)$	$\Rightarrow I, [3 : 7]$
9	$\forall x.\ (S(x) \Rightarrow P(x))$	$\forall I, [8]$

Claim Given $\forall x.\exists y.\ L(x,y)$ and $\forall x.\forall y.\ (L(x,y) \Rightarrow F(x,y))$, prove that $\forall x.\exists y.\ F(x,y)$, where $L(x,y)$: x likes y and $F(x,y)$: x is friends with y.

Let x be an arbitrary person.

1	$\forall x.\exists y.\ L(x,y)$	*Premise*
2	$\forall x.\forall y.\ (L(x,y) \Rightarrow F(x,y))$	*Premise*
3	$\exists y.\ L(x,y)$	$\forall E, [1]$
4	$a, L(x,a)$	$\exists E, [3]$
5	$\forall y.\ (L(x,y) \Rightarrow F(x,y))$	$\forall E, [2]$
6	$L(x,a) \Rightarrow F(x,a)$	$\forall E, [5]$
7	$F(x,a)$	$\Rightarrow E, [4:6]$
8	$\exists y.\ F(x,y)$	$\exists I, [7]$
9	$\forall x.\exists y.\ F(x,y)$	$\forall I, [8]$

Claim: Given $\forall x.\ (P(x) \Rightarrow S(x))$, $\exists x.\ P(x)$, and $\forall x.\ (S(x) \Rightarrow \exists y.\ R(x,y))$, prove $\exists x.\exists y.\ R(x,y)$, where $P(x)$: x is a professor; $S(x)$: x is a scientist; $R(x,y)$: x researches y.

Let x be an arbitrary person.

1	$\forall x.\ (P(x) \Rightarrow S(x))$	*Premise*
2	$\exists x.\ P(x)$	*Premise*
3	$\forall x.\ (S(x) \Rightarrow \exists y.\ R(x,y))$	*Premise*
4	$a, P(a)$	$\exists E, [2]$
5	$P(a) \Rightarrow S(a)$	$\forall E, [1]$
6	$S(a)$	$\Rightarrow E, [4,5]$
7	$S(a) \Rightarrow \exists y.\ R(a,y)$	$\forall E, [3]$
8	$\exists y.\ R(a,y)$	$\Rightarrow E, [6,7]$
9	$b, R(a,b)$	$\exists E, [8]$
10	$\exists y.R(a,y)$	$\exists I, [9]$
11	$\exists x.\exists y.R(x,y)$	$\exists I, [10]$